Maxwell Morrison

Electrical Engineering and Applied Mathematics Student

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maxwellmorrison.net

University of Delaware student pursuing electrical engineering and applied mathematics, with interests in signal processing, embedded systems, machine learning, computational neural information engineering, and practical engineering tools.

EDUCATION

University of Delaware
Expected May 2027

Newark, DE

- B.S. Electrical Engineering, College of Engineering
- B.S. Applied Mathematics, College of Arts and Sciences
- GPA: 3.6

Dulaney High School
May 2023

Timonium, MD

- High School Diploma

RESEARCH, TEACHING, AND ENGINEERING EXPERIENCE

Undergraduate Researcher
Summer 2026

Computational Neural Information Engineering Laboratory (CNIEL), University of Delaware | Newark, DE

- Conducting undergraduate research under Dr. Austin Brockmeier in the Computational Neural Information Engineering Laboratory.

Supplemental Instructor / Teaching Assistant
Jan. 2025 - Present

University of Delaware | Newark, DE

- Supplemental Instructor for CPEG202, Introduction to Digital Logic, under Dr. Nathan Lazarus and Mr. Thomas Lum.
- Hold weekly student support hours and assist students with course concepts, lab work, and problem solving.
- Support instruction through grading quizzes, exams, and course materials while serving as a primary contact for students.

Lab Assistant
Jan. 2025 - Present

University of Delaware | Newark, DE

- Lab Assistant for ELEG205, Introduction to Analog Circuits, under Dr. Fouad Kiamilev.
- Lead weekly lab sessions and support students in circuit analysis, measurement techniques, and lab procedures.
- Assist with grading lab coursework and help ensure smooth operation of instructional lab activities.

Electrical Engineering Intern
May 2025 - Aug. 2025

Henry Adams | Baltimore, MD

- Worked in Revit and AutoCAD to develop detailed drawings for contractors.
- Performed lighting simulations using AGi-32.
- Contributed to arc flash studies using SKM.

Undergraduate Researcher
Jan. 2025 - Aug. 2025

University of Delaware | Newark, DE

- Worked with Dr. Mohsen Badiey and a team of undergraduate researchers on an electronic acoustic recorder.
- Developed a system using the ESP-WROOM-32D with custom hardware and firmware.
- Contributed to embedded system development in support of underwater acoustics research.

PROJECTS

Drone Eye

Senior Design, Computer Vision, LiDAR

Computer-vision and machine-learning drone detection system using LiDAR for depth perception and a custom-trained model to detect drones and estimate Cartesian coordinates through a graphical interface.

Diabeasy

UD Hackathon, Python

Encrypted Tkinter application for tracking insulin usage and calculating dosage amounts.

PitchMAX

Hardware, Firmware

Lower-cost pitch calling system inspired by PitchCom, using custom hardware and firmware to communicate called pitches between players.

PyroPlan

UD Hackathon, Web App, Simulation

Web platform for modeling fire spread in level 1 materials server-side, making simulation accessible without local software installation.

Mom Blood Sugar Tracker

Healthcare, Web App

Web platform that lets multiple caregivers submit orders and coordinate blood sugar and insulin-related care.

AWARDS

- 2026 Research Day People's Choice Award Winner for DroneEye, Department of Electrical and Computer Engineering, University of Delaware
- Institute of Electrical and Electronics Engineers Delaware Bay Section Student Activities Award, Department of Electrical and Computer Engineering, University of Delaware
- Eagle Scout, 2021
- Dean's List

AFFILIATIONS

- IEEE Student Chapter
- IEEE Eta Kappa Nu
- Troop 97 (Scouts BSA)

TECHNICAL SKILLS

- Signal processing
- Machine learning
- Embedded systems
- Digital logic
- Analog circuits
- Python
- Tkinter
- Revit
- AutoCAD
- AGi-32
- SKM
- ESP-WROOM-32D